

Optics Letters

Coherence enhancement of a chirped DFB laser for frequency-modulated continuous-wave reflectometry using a composite feedback loop

JIE QIN, QIAN ZHOU, WEILIN XIE, YAN XU, SHENGGUO YU, ZHANGWEIYI LIU, YI TIAN TONG, YI DONG,* AND WEISHENG HU

State Key Laboratory of Advanced Optical Communication Systems and Networks, Shanghai Jiao Tong University, Shanghai 200240, China

*Corresponding author: yidong@sjtu.edu.cn

Received 3 July 2015; revised 14 August 2015; accepted 1 September 2015; posted 2 September 2015 (Doc. ID 244230); published 28 September 2015

We demonstrate efficient coherence enhancement of a chirped distributed feedback (DFB) laser for frequency-modulated continuous-wave (FMCW) reflectometry. Both sweep nonlinearity and broadband stochastic frequency noises during the laser chirp are efficiently suppressed by a composite feedback loop. The residual frequency error relative to a perfect linear chirp is shown to be about 89 kHz for a laser chirp of 50 GHz in 100 ms, compared with 44 MHz with the loop open. The broadband frequency noise suppression of the frequency-swept laser greatly improves its coherence, leading to a higher signal-to-noise ratio and a significantly extended measurement range in FMCW reflectometry ranging. We demonstrate a 2 mm transform-limited spatial resolution at a range window of 50 m and a 17.5 cm spatial resolution at an extended measurement range of 750 m, which is about 15 times the intrinsic laser round-trip coherence length. © 2015 Optical Society of America

OCIS codes: (140.3518) Lasers, frequency modulated; (120.3180) Interferometry; (140.3600) Lasers, tunable; (140.3490) Lasers, distributed-feedback.

<http://dx.doi.org/10.1364/OL.40.004500>

Optical frequency-modulated continuous-wave (FMCW) reflectometry is a promising candidate for analysis of local area networks, as well as for characterizing fiber-optic components such as fiber connectors, packaged optical devices, and optical integrated devices [1–5]. Lasers with high spectral purity and a large tuning range are important for coherent FMCW reflectometry, as the maximum range is limited by the coherence length of the laser and the spatial resolution is inversely proportional to the frequency tuning span. However, the simultaneous achievement of low phase noise operation and precise broadband tuning is challenging for conventional laser sources [6]. Then for FMCW reflectometry ranging measurement, there is usually a trade-off between the spatial resolution and the maximum available ranging window [7,8].

For tunable laser sources used in FMCW reflectometry, sweep nonlinearity and phase noise performance are two important metrics. Sweep nonlinearity corresponds to the nonuniform frequency modulation (FM) response of the frequency tuning factors, such as injection current or mechanical tuning of the laser cavity [9,10]; it is usually of the order of the repetition frequency of the waveform, and, thus, is a deterministic low frequency noise component. Phase noise is related to the phase random walk induced by the $1/f$ frequency noise and the spontaneous emission; it is a broadband stochastic frequency noise component [11]. Generally, the spatial resolution of the FMCW reflectometry is severely degraded as sweep nonlinearity exists during the laser chirp, while the maximum range window is limited because of large phase noise [12]. Tunable fiber lasers intrinsically have a narrow linewidth and low phase noise operation, but their small tuning range and inherent tuning nonlinearity limit their applications in high spatial resolution FMCW reflectometry. In addition, fiber lasers have a relatively large volume and high power consumption, as well as a high cost. In Ref. 13, an experiment based on the servo-loop control method using self-heterodyning delay interferometers [13–19] is carried out for nonlinearity frequency error correction and frequency stabilization in low Fourier frequencies. However, the loop bandwidth is limited by the long fiber delay line in the interferometer and slow response time of the acousto-optic modulator, in which case the loop is not able to suppress broadband frequency noise. Compared with fiber lasers, distributed feedback (DFB) lasers are more compact, less expensive, and inherently more efficient. DFB lasers have been attractive candidates for FMCW reflectometry for a long time. However, DFB lasers exploit inherent sweep nonlinearity and large phase noise [7–9,20]. To date, the difficulty related to the application of these lasers in coherent FMCW reflectometry is the degradation of spatial resolution because of sweep nonlinearity, and the limited measurement range because of the short coherence length [7–9,12]. However, the available bandwidth of servo-loop control by injecting current to the semiconductor laser is limited to several hundred kHz by the phase reversal of the laser FM response [21]; thus, the improvement for the laser

coherence is constrained. As a result, the measurement range of coherent FMCW reflectometry based on DFB laser servo-controlled simply by injection current is limited to a short range [19].

In this Letter, we propose a composite feedback loop consisting of two feedback loops, one that is used to control the DFB laser's injection current mainly to compensate for sweep nonlinearity, and the other to control the frequency of a specially customized acousto-optic frequency shifter (AOFS) with a fast response time to compensate for broadband stochastic frequency noise. The sweep nonlinearity suppression and broadband frequency noise reduction of the frequency-swept DFB laser greatly improve its coherence, leading to a higher signal-to-noise ratio and a significantly extended measurement range in FMCW reflectometry ranging.

The experimental setup for sweep nonlinearity and broadband frequency noise suppression is presented in Fig. 1. The frequency-swept laser used in the experiment is a commercially available fiber-coupled DFB laser (~ 625 kHz linewidth at -3 dB) at a wavelength of 1550 nm, and frequency tuning is realized by injection current tuning. The output of the laser is first passed through an acousto-optic frequency shifter (AOFS1) operating at 500 MHz. Then the laser is split into two parts; a 90% portion of the output is used for FMCW ranging experiments. The 10% part is coupled into a frequency-shifted Mach-Zehnder (MZ) interferometer with an acousto-optic frequency shifter (AOFS2) operating at 40 MHz driven by a stable RF signal in one arm and an optical fiber delay line in another arm. The delayed beam and the frequency shifted beam are recombined, and the heterodyne beat signal is detected by a photo detector. Polarization maintaining fiberoptic components are used within the composite feedback loop to avoid signal fading because of polarization misalignment.

The frequency-shifted Mach-Zehnder interferometer is used to transfer the laser frequency fluctuations $\delta\nu$ to heterodyne signal phase fluctuations $\delta\phi$ [13,16]; the transfer function is

$$\frac{\delta\phi(f)}{\delta\nu(f)} = (1 - e^{-i2\pi f\tau_0})/if \approx 2\pi\tau_0 \quad (1)$$

$$f \ll 1/\tau_0 \text{ (rad/Hz),}$$

where τ_0 is the fiber delay time and f is the Fourier frequency. For low Fourier frequencies ($f \ll 1/\tau_0$), the magnitude of the transfer function is approximately equal to $2\pi\tau_0$. Here, we choose a time delay of 45.5 ns; experimental results show that the time delay is adequate enough for sweep nonlinearity

suppression and broadband stochastic frequency noise reduction as well. Assume that the time domain instantaneous optical frequency of the chirped laser is expressed as

$$\nu(t) = \nu_0 + rt + \delta\nu(t), \quad (2)$$

where ν_0 is the initial optical frequency, r is the tuning rate of the optical frequency sweep, and $\delta\nu(t)$ corresponds to the frequency errors during the laser chirp. The beat note signal at the output of the interferometer can be expressed as

$$I_{PD}(t) = I_0 \cos \left(2\pi f_b t + \varphi_0 + 2\pi \int_{t-\tau_0}^t \delta\nu(\xi) d\xi \right), \quad (3)$$

where I_0 is the beat note amplitude, $f_b = r\tau_0 + f_{\text{AOFS2}}$ is the beat note frequency with $f_{\text{AOFS2}} = 40$ MHz the driving frequency of the AOFS2, and φ_0 is a constant phase term. Consider that the propagation delay is small compared to the typical variation time scale of $\delta\nu(t)$, the beat note phase term can be expressed as

$$\phi(t) = 2\pi f_b t + \varphi_0 + 2\pi\tau_0\delta\nu(t). \quad (4)$$

The phase error term is $\delta\phi(t) = 2\pi\tau_0\delta\nu(t)$. It is clear that the beat note phase error is well proportioned to the frequency error during the laser chirp. The phase and frequency of this beat note are compared to a stable RF reference signal provided by an arbitrary/function generator (Tektronix AFG3252) at frequency $f_{\text{ref}} = r\tau_0 + f_{\text{AOFS2}}$ using a low-noise digital phase frequency detector (DPFD) to provide an error signal proportional to $\delta\phi(t)$. The error signal is then split into two parts. One part is amplified, filtered, and fed back to the laser current controller; the loop filter consists of a second-order loop filter with an integration stage to 30 kHz and a low-pass pole of 2.6 MHz, and the feedback is mainly to correct the low-frequency, but large scale sweep nonlinearity. The other part is amplified, filtered by a low-pass filter with a cutoff frequency of 4 MHz and fed back to an acousto-optic frequency shifter (AOFS1) for broadband stochastic frequency error correction; it should be pointed out that the available loop bandwidth is directly related to the response time of AOFS1 and the AOFS1 used in our system which is specially customized to guarantee a fast response time (~ 65 ns).

We first measure the linewidths of the laser under non-chirped operation using a delayed self-heterodyne interferometer. The linewidths under single loop (i.e., the current feedback loop is closed, and the AOFS feedback loop is open) and composite loop operations are 350 and 150 kHz, respectively. To evaluate the frequency noise suppression ability of the servo-loop, the phase noise of the 40 MHz beat note signal at the output of PD1 is measured by a phase noise analyzer (Symmetricom 5125A), which is then converted into laser frequency noise PSD using the scaling factor $1/(2\pi\tau_0)^2$. The frequency noise PSD is plotted in Fig. 2; the presence of the strong peaks is mainly attributed to laser drive current noise. When the loop is open, one can see a quasi- $1/f$ noise. The noise level is suppressed in a 1 Hz to 500 kHz frequency range under single-loop operation. In addition, the noise level is further suppressed to 3–5 dB within a ~ 1 MHz frequency range under composite loop operation. It is obvious that the composite feedback loop configuration allows better frequency error suppression and enables a much larger loop control bandwidth.

As for frequency sweep, the laser frequency-current tuning coefficient is averaged to be about 0.417 GHz/mA at a center

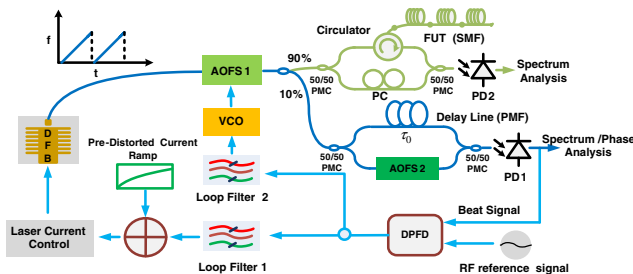


Fig. 1. Schematic of the experimental setup. AOFS, acousto-optic frequency shifter; FUT, fiber under test; SMF, single-mode fiber; PMC, polarization-maintaining coupler; PC, polarization controller; PMF, polarization-maintaining fiber; PD, photo detector; DPFD, digital phase frequency detector; VCO, voltage controlled oscillator.

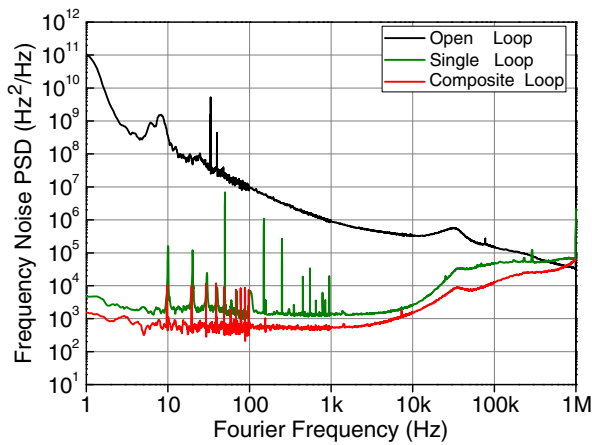


Fig. 2. PSD of the laser frequency noise under nonchirped laser operation. Black line, free running laser; green line, single current feedback loop operation; red line, composite feedback loop operation.

wavelength of 1550 nm. We first apply a predistortion scheme to remove as much of the systematic chirp nonlinearities as possible; using a Hilbert transform technique demonstrated by Ahn and Kim [22], we measured frequency chirp nonlinearities and applied a time-varying current ramp designed to produce a linear optical frequency sweep to the laser current controller. The current tuning range is 80–200 mA in a 100 ms time period, corresponding to a ~ 50 GHz optical frequency tuning span. The chirp spectral purity is tested by Fourier analysis of the detected heterodyne beat signal at the output of PD1 using a high-performance real-time signal analyzer (Tektronix RSA5103A). A trigger signal synchronized to the current ramp signal is used for data acquisition, and a Hanning window is used in the Fourier transform process to minimize the side lobe effects. We could see that the frequency of the beat note is shown to be 40.023 MHz (note that $f_b = r\tau_0 + f_{\text{AOFSS2}}$), corresponding to an optical frequency sweep rate of 500 GHz/s tuning rate. The spectra of the detected photo current beat signal at the output of PD1 when the loop is open, when the single current feedback loop is closed, and when the composite

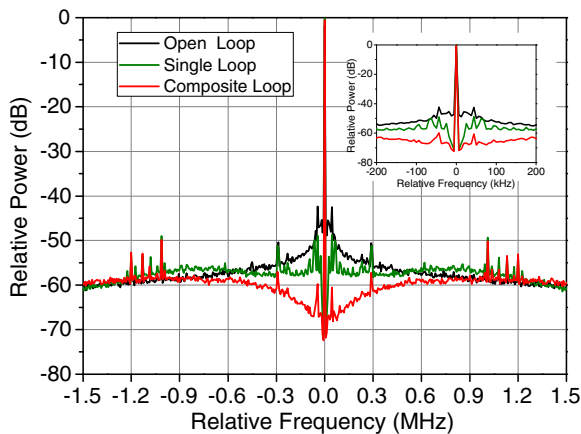


Fig. 3. Beat note spectrum when the laser frequency is chirped over 50 GHz in 100 ms. Black line, open loop operation; green line, single feedback loop operation; red line, composite feedback loop operation.

feedback loop is closed are shown in Fig. 3 (relative to the beat note frequency). When the loop is open, although variation of the chirp rate with time is largely reduced with the application of the predistortion, the beat note spectrum still exhibits a large noise pedestal because of residual sweep nonlinearity and phase noise. In comparison, when the feedback loop is closed, the residual sweep nonlinearity and the phase noise within the loop control bandwidth are efficiently suppressed. Compared with single feedback loop operation, the noise floor of the beat note signal is further rejected in composite loop operation, and the loop control bandwidth is extended to MHz level.

We then assess the frequency error during the whole laser chirp by analyzing the time domain beat note phase error. The time domain beat note phase term $\phi(t) = 2\pi f_b t + \varphi_0 + \delta\varphi(t)$ is recorded and digitalized at the output of PD1 using the phase versus time function of the real-time signal analyzer. Removing the linear term $2\pi f_b t + \varphi_0$, the residual beat note phase error term $\delta\varphi(t)$ is extracted. The beat note phase error is converted into optical frequency error using the scale $1/(2\pi\tau_0)$, and then the frequency error is obtained. As shown in Fig. 4(a), when the loop is open, the maximum laser frequency deviation from a linear chirp is about 44 MHz. In comparison, the green and the red curves in Fig. 4(b) correspond to the cases when the single feedback loop and the composite feedback loop are closed, respectively. The standard deviation of frequency error in single feedback loop operation is 153 kHz, while that of composite feedback loop operation is shown to be only 89 kHz; thus, the composite feedback loop gives a better frequency error suppression ability than the single-loop configuration.

The influence of the broadband frequency noise suppression of the frequency-swept DFB laser in FMCW reflectometry ranging is carried out with an optical frequency domain reflectometry (OFDR) interferometric experiment. Light from the

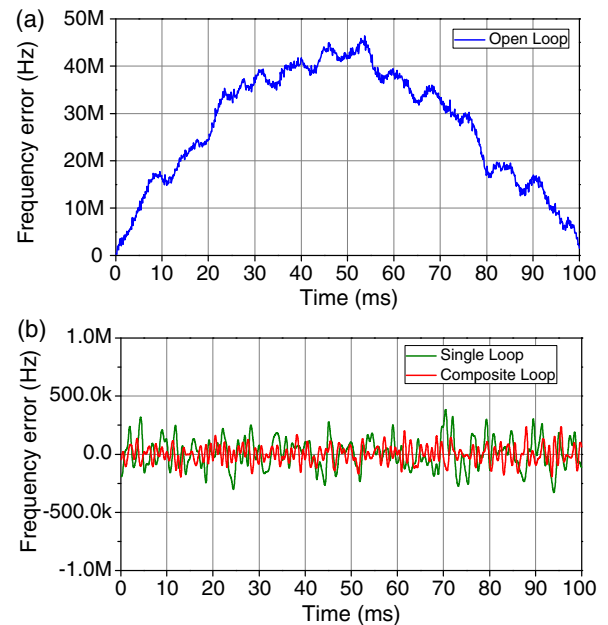


Fig. 4. (a) Frequency error relative to linearity for a 50 GHz frequency chirp in 100 ms under open loop operation. (b) Frequency error under closed loop operation. Green line, single feedback loop operation; red line, composite feedback loop operation.

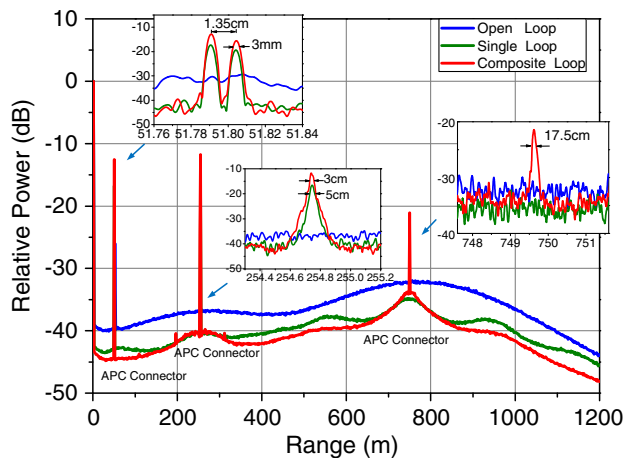


Fig. 5. FMCW reflectometry ranging results using an OFDR experimental setup. Blue line, open loop operation; green line, single feedback loop operation; red line, composite feedback loop operation.

frequency-swept DFB laser source is split between the reference and measurement arms of a MZ interferometer. The beat signal of the reference and the measurement fields is then detected by a photo detector, and multiple reflection events are visualized using Fourier analysis (10 times the average with a Hanning window). The FUT consists of three spools of standard single-mode fiber with lengths of 50, 200, and 500 m, connected by an APC fiber connector; a bad splice between the 50 m fiber end and the APC connector with a space of 1.35 cm is designed to measure the ability of the linearly swept DFB laser to resolve closely spaced targets. Results of the measurement are shown in Fig. 5. At the distance of 50 m, one could see a reflection event in open loop operation, but the reflection peaks are not able to be localized and distinguished because of sweep nonlinearity and phase noise. In contrast, the reflection peaks could be precisely localized, and the two closely spaced reflection peaks could be clearly resolved in close loop operation because of the bad splice and the APC connector. The full width at half-maximum (FWHM) of the reflection peak is shown to be 3 mm (2 mm without a Hanning window), which is the transform-limited resolution of the 50 GHz tuning span. As far as we know, it is the longest range window, while the transform-limited resolution is maintained when an injection current tuned DFB laser is used as the frequency-swept light source. At the distance of 250 m, the reflection peak in open loop operation is totally submerged in the noise floor because of sweep nonlinearity and large phase noise. In contrast, the reflection peaks could be precisely localized in close loop operation; the spatial resolution based on FWHM are shown to be 5 and 3 cm in single feedback loop and composite feedback loop operation; the spatial resolution degradation relative to theoretical resolution is because of the presence of residual flicker noise according to our study. One could see the great advantage of the proposed composite feedback loop more obviously in a long range circumstance. At the distance of 750 m, which is about 15 times the round-trip coherence length (~ 50 m) of the DFB laser, reflection peaks in both open loop and single feedback loop operation are submerged in the phase noise floor and totally invisible, while one could still see a clear reflection peak with a spatial resolution of 17.5 cm in composite feedback

loop operation. The extended measurement range indicates a significant coherence enhancement of the chirped laser source because of the broadband frequency noise suppression ability of the proposed composite feedback loop.

In summary, we demonstrate efficient coherence enhancement of a chirped DFB laser using a composite feedback loop. Both sweep nonlinearity and broadband stochastic frequency noise up to MHz level are effectively suppressed by the loop. For a laser chirp of 50 GHz in 100 ms, the residual frequency error relative to a perfect linear sweep is shown to be about 89 kHz, compared with 44 MHz with the loop open. The broadband frequency noise suppression of the frequency-swept DFB laser greatly improves its coherence, leading to a higher signal-to-noise ratio and an extended measurement range in the FMCW reflectometry ranging experiment. We demonstrate a 2 mm transform-limited spatial resolution at a measurement range of 50 m, a 3 cm spatial resolution at a measurement range of 250 m, and a 17.5 cm spatial resolution at an extended measurement range of 750 m, which is 15 times that of the intrinsic laser round-trip coherence length.

Funding. 973 Program (2012CB315602); National Natural Science Foundation of China (NSFC) (61225004).

REFERENCES

1. L.-T. Wang, K. Iiyama, F. Tsukada, N. Yoshida, and K. Hayashi, *Opt. Lett.* **18**, 1095 (1993).
2. U. Glombitza and E. Brinkmeyer, *J. Lightwave Technol.* **11**, 1377 (1993).
3. T.-J. Ahn, S. Moon, Y. Youk, Y. Jung, K. Oh, and D. Y. Kim, *Opt. Express* **13**, 4005 (2005).
4. B. J. Soller, D. K. Gifford, M. S. Wolfe, and M. E. Froggatt, *Opt. Express* **13**, 666 (2005).
5. K. Iiyama, S.-I. Matsui, T. Kobayashi, and T. Maruyama, *IEEE Photon. Technol. Lett.* **23**, 703 (2011).
6. W. Demtroder, *Laser Spectroscopy* (Springer, 1996).
7. J. P. von der Weid, R. Passy, and N. Gisin, *J. Lightwave Technol.* **13**, 954 (1995).
8. P. Oberson, B. Huttner, O. Guinnard, L. Guinnard, G. Ribordy, and N. Gisin, *IEEE Photon. Technol. Lett.* **12**, 867 (2000).
9. R. Passy, N. Gisin, J. P. von der Weid, and H. H. Gilgen, *J. Lightwave Technol.* **12**, 1622 (1994).
10. C. J. Karlsson and F. A. A. Olsson, *Appl. Opt.* **38**, 3376 (1999).
11. G. Gorju, V. Crozatier, V. Lavielle, I. Lorg  r  , J.-L. Le Gou  t, and F. Bretenaker, *Eur. Phys. J. Appl. Phys.* **30**, 175 (2005).
12. J. P. von der Weid, R. Passy, G. Muss, and N. Gisin, *J. Lightwave Technol.* **15**, 1131 (1997).
13. H. Jiang, F. K  f  lian, P. Lemonde, A. Clairon, and G. Santarelli, *Opt. Express* **18**, 3284 (2010).
14. K. Iiyama, L.-T. Wang, and K.-I. Hayashi, *J. Lightwave Technol.* **14**, 173 (1996).
15. C. Greiner, B. Boggs, T. Wang, and T. Mossberg, *Opt. Lett.* **23**, 1280 (1998).
16. V. Crozatier, G. Gorju, F. Bretenaker, J.-L. Le Gou  t, I. Lorg  r  , C. Gagnol, and E. Ducloux, *Appl. Phys. Lett.* **89**, 261115 (2006).
17. G. Gorju, A. Jucha, A. Jain, V. Crozatier, I. Lorg  r  , J.-L. Le Gou  t, F. Bretenaker, and M. Colice, *Opt. Lett.* **32**, 484 (2007).
18. P. A. Roos, R. R. Reibel, B. Kaylor, T. Berg, Z. W. Barber, and W. R. Babbitt, *Opt. Lett.* **34**, 3692 (2009).
19. N. Satyan, A. Vasilyev, G. Rakuljic, V. Leyva, and A. Yariv, *Opt. Express* **17**, 15991 (2009).
20. C. Henry, *IEEE J. Quantum Electron.* **18**, 259 (1982).
21. P. Corrc, O. Girad, and I. F. de Faria, Jr., *IEEE J. Quantum Electron.* **30**, 2485 (1994).
22. T.-J. Ahn and D. Y. Kim, *Appl. Opt.* **46**, 2394 (2007).